SmartRFP

PRODUCT REQUIREMENTS DOCUMENT · CAPSTONE PROJECT

SmartRFP — agentic AI that drafts your RFP responses in minutes, not days.

An autonomous system that reads an incoming RFP, pulls the right knowledge from your enterprise library, fetches live pricing, and writes a structured draft response — then hands it to a human to approve, edit, or send back.

PROJECT TYPE
Agentic AI / RAG Application

ORCHESTRATION
LangGraph (multi-agent)

CONTROL PATTERN
Human-in-the-Loop

PRIMARY USERS
Presales · Bid Managers · SMEs

01 The problem we are solving

When a client sends an RFP, the response team does the same slow, manual loop every single time. SmartRFP automates the repetitive 80% so people spend their time on the high-value 20%.

Slow turnaround

Reading requirements, hunting old proposals, and chasing prices can take a team several days per RFP.

↻

Inconsistent answers

Different people write differently. Quality and tone vary from bid to bid, and good past answers get rewritten from scratch.

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Outdated pricing

Teams copy old numbers from last year's deck. Margins and SKU prices drift, and the proposal goes out wrong.

⌥

Low knowledge reuse

Years of winning proposals sit in folders nobody searches. The same wheel is reinvented every quarter.

⚑

Compliance gaps

Mandatory clauses and certifications get missed under deadline pressure, risking disqualification.

⚇

Heavy coordination

Every response depends on pulling in many internal teams, creating bottlenecks and dependency.

02 Goals & non-goals

Goals — what SmartRFP will do

- Automatically parse an uploaded RFP and break it into clear, answerable requirements.
- Retrieve the most relevant past answers, templates, and compliance clauses using RAG.
- Fetch real-time pricing and cost data from live APIs at the moment of drafting.
- Generate a structured, section-by-section draft response with sources cited.
- Flag compliance risks, missing information, and likely hallucinations for the reviewer.
- Give a human reviewer a clear screen to approve, edit, or regenerate before anything is final.
- Cut average draft turnaround from days to under an hour.

Non-goals — what it will NOT do (for now)

- It will not auto-submit responses to clients. A human always approves first.
- It will not replace presales, bid managers, or SMEs — it assists them.
- It will not negotiate, set final margins, or make commercial sign-off decisions.
- It will not handle e-signatures, contracts, or legal execution of the deal.
- It will not invent pricing if a live API is unavailable — it flags the gap instead.
- It will not manage project delivery after the bid is won.
- Multi-language translation is out of scope for the first version.

03 Who uses it

BM

Bid Manager

Owns the response. Uploads the RFP, reviews the full draft, manages the approval gate, and exports the final document.

PS

Presales Engineer

Checks technical answers and the solution design pulled from past proposals. Edits sections that need expert nuance.

SME

Subject Matter Expert

Gets pinged only for flagged sections. Validates compliance clauses, pricing assumptions, and risky claims.

04 Feature definitions

Six core features make up the product. The two agents (F2 and F3) run **in parallel** after parsing, which is what makes drafting fast.

Automation / AI   Human-in-the-loop

F1 Intake

RFP Parser & Requirement Extractor

Accepts PDF / DOCX uploads. Extracts text, splits the document into sections, and turns it into a clean list of requirements and questions to be answered.

F2 Agent 1 Parallel

RAG Retrieval Agent

Searches the enterprise knowledge base — past RFP answers, templates, solution docs, compliance clauses, product catalog, case studies — and returns the most relevant content per requirement, with sources.

F3 Agent 2 Parallel

Live Pricing & Cost Intelligence Agent

Calls external/internal APIs for current component cost, service pricing, labor rates, SKU prices, regional pricing, and tax/margin inputs — so commercials are current, not copied from last year.

F4 Synthesis

Draft Generator

A synthesis node merges both agents' outputs into a structured draft: technical answers + commercial table + compliance section, written in a consistent house style.

F5 HITL

Review & Approval Workspace

The reviewer sees the draft with highlighted pricing assumptions, compliance flags, missing-info markers, and hallucination-risk warnings. They can **approve, edit inline, or send back for regeneration.**

F6 Output

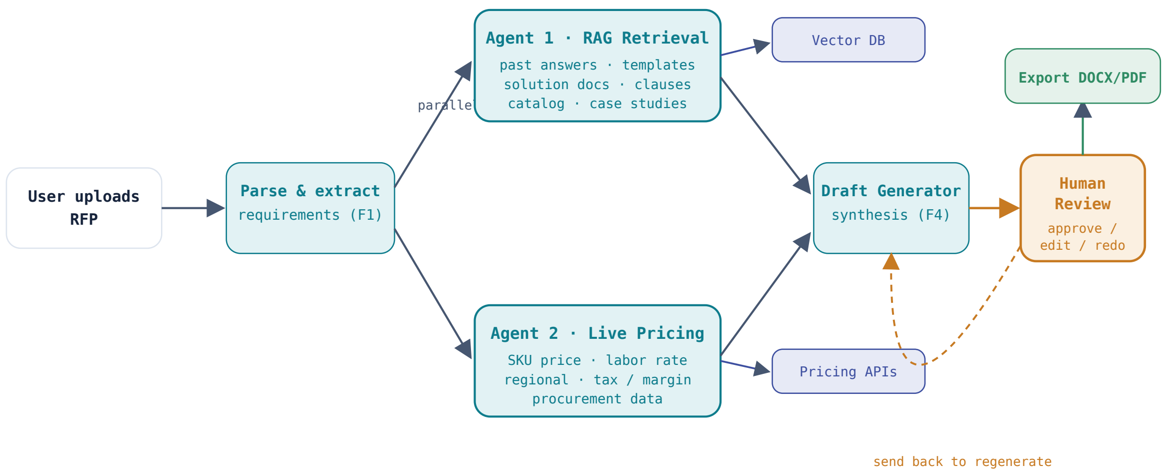
Export & Audit Trail

Exports the approved response to DOCX/PDF and keeps a record of what the AI suggested, what the human changed, and which sources and prices were used.

05 System architecture

LangGraph is the orchestration engine. It runs the graph below: parse → fork into two parallel agents → join → synthesize → pause at the human gate → finalize.

AI / automated node   Human gate   Data / tools



LANGGRAPH ORCHESTRATION — PARSE → PARALLEL AGENTS → SYNTHESIZE → HUMAN GATE → EXPORT

06 How it works — start to finish

The full journey from a fresh RFP in the inbox to an approved, exported response.

Upload the RFP

The bid manager drops the client's RFP (PDF or DOCX) into SmartRFP and confirms the deal name, region, and deadline.

USER ACTION

Parse & extract requirements

The system reads the document, splits it into sections, and produces a structured checklist of every requirement and question that needs an answer.

F1 · AUTOMATED

Two agents run in parallel

LangGraph forks the workflow. **Agent 1** retrieves matching knowledge from the enterprise library. **Agent 2** calls live pricing APIs. Running together keeps it fast.

F2 + F3 · PARALLEL

Synthesize the draft

Outputs from both agents join at the synthesis node, which writes a structured draft: technical answers, a current pricing table, and a compliance section — all in one consistent voice, with sources attached.

F4 · AUTOMATED

Human review gate (the workflow pauses here)

The reviewer opens the workspace and sees the draft with pricing assumptions, compliance flags, missing-info markers, and hallucination-risk warnings highlighted. Nothing moves forward without them.

F5 · HUMAN-IN-THE-LOOP

Approve, edit, or send back

Approve → continue to export. **Edit** → make inline changes, then approve. **Send back** → the workflow loops to regenerate weak sections with reviewer notes as guidance.

F5 · HUMAN DECISION

Export & log

The approved response exports to DOCX/PDF. An audit trail records the AI suggestions, human edits, sources, and prices used — ready for the next bid.

F6 · AUTOMATED

07 UI mockups

Three key screens: the intake screen, the live generation screen showing both agents working, and the human review workspace where the real decisions happen.

Screen 1 · RFP intake

smartrfp.app / new-response

Start a new RFP response

Upload the client document. We'll read it and pull out every requirement automatically.

⤓ Drop RFP here or click to upload

PDF or DOCX · up to 50 MB

DEAL NAME

Acme Cloud Migration — RFP-2026-118

REGION

North America

DEADLINE

10 Jul 2026

Parse & generate draft → **Save for later**

Screen 2 · Generating (agents working in parallel)

smartrfp.app / RFP-2026-118 / generating

Building your draft

42 requirements found. Both agents are running together.

Agent 1 · RAG Retrieval

Matching requirements to past proposals & compliance clauses…

38 / 42 sections matched

Agent 2 · Live Pricing

Fetching current SKU prices, labor rates & regional margin…

7 / 9 cost items priced

Synthesizing draft · ~25s remaining

Screen 3 · Human review workspace (the approval gate)

smartrfp.app / RFP-2026-118 / review

Draft ready for review

Acme Cloud Migration · 42 sections · 3 flags need attention

✓ Approve ✎ Edit ↩ Send back

2.1 Solution Overview

We propose a phased migration to a managed cloud platform with 99.95% uptime SLA, drawn from our prior Contoso migration.

4.0 Commercial Summary

Estimated total: $486,200 based on current SKU pricing pulled today.

6.2 Compliance — Data Residency

Customer data will remain in-region per local regulation. (clause needs SME confirmation)

● Pricing — live

Compute SKU x12	**$214,000**
Labor (480 hrs)	**$192,000**
Margin 18%	**$80,200**

fetched 25 Jun 2026 · source: pricing-api

⚑ Compliance flag

Data-residency clause auto-filled from template. **SME must confirm** it matches this client's region.

⚠ Hallucination risk

The 99.95% SLA claim is **not found in any source doc**. Verify before sending. `low confidence`

○ Missing info

Section 8 (References) is empty — no matching case study found.

Screen 4 · Dashboard

smartrfp.app / dashboard

Your RFP responses

DEAL	DEADLINE	FLAGS	STATUS
Acme Cloud Migration	10 Jul	3	In review
Globex Security Audit	14 Jul	0	Drafting
Initech Data Platform	28 Jun	0	Approved

+ New response

08 Real-time scenarios

Three ways the system behaves in practice — including what happens when things go wrong.

SCENARIO A · THE HAPPY PATH

A 40-page cloud RFP, drafted in under 30 minutes

A bid manager uploads Acme's RFP at 9:00 AM. By 9:25 a full draft is ready with current pricing. Two flags appear; the SME confirms the data-residency clause and one SLA number. Approved and exported by 10:15 — a job that used to take three days.

Upload → Parse 42 reqs → RAG + Pricing (parallel) → Draft → 2 flags resolved → Export

SCENARIO B · A FLAG SAVES THE DEAL

The pricing API is stale, and SmartRFP catches it

Agent 2 calls the pricing API but a SKU returns a timestamp from last quarter. Instead of using the old number, SmartRFP marks the line "stale — verify" in red and does not finalize the commercial table. The reviewer refreshes the price before the proposal goes out, avoiding a margin error.

Pricing API timeout → Mark line stale → Block auto-finalize → Human refreshes

SCENARIO C · SEND BACK TO REGENERATE

The reviewer rejects a weak technical section

The presales engineer finds section 3 too generic. They click **Send back** and add a note: "use the Contoso reference architecture." LangGraph loops only that section through the RAG agent again with the note as guidance, regenerates it, and returns it for a second review — without redoing the whole document.

Reviewer note → Loop section 3 → RAG re-run → Re-review

09 How it performs

Targets for the first version. The two agents running in parallel is the single biggest speed lever.

< 30min Upload → reviewable draft	**~50%** Faster from parallel agents vs. sequential
100% Drafts pass through a human gate	**≥ 85%** Retrieved sections rated relevant

Speed & reliability

- Parallel execution: RAG and Pricing agents never wait on each other.
- If a pricing API is slow or down, the workflow continues and flags the gap rather than stalling.
- Regeneration re-runs only the affected sections, not the whole document.
- Caching of recent price lookups to reduce repeat API calls and latency.

Quality & trust

- Every retrieved answer carries its source so reviewers can verify quickly.
- Claims with no supporting source are flagged as hallucination risk.
- Pricing lines show a fetch timestamp and source API.
- Full audit trail of AI suggestions vs. human edits for every response.

10 Test cases

Coverage across functional, performance, human-in-the-loop, and edge/failure cases.

Functional   Performance   HITL   Edge / failure

ID	TYPE	TEST	EXPECTED RESULT
TC-01	Func	Upload a valid 30-page PDF RFP.	Document parsed; requirement checklist generated with no sections lost.
TC-02	Func	Upload a DOCX with tables and images.	Text extracted correctly; tables read; images skipped without crashing.
TC-03	Func	RAG agent queries a requirement with a known matching past answer.	Correct past answer retrieved and ranked top, with source shown.
TC-04	Func	Pricing agent requests a known SKU price.	Current price returned with timestamp and source API tag.

TC-05	Func	Synthesis node combines both agent outputs.	Draft contains technical answers, a pricing table, and a compliance section in one consistent format.
TC-06	Perf	Confirm RAG and Pricing agents run in parallel.	Total time ≈ the slower agent, not the sum of both.
TC-07	Perf	Generate a draft for a 40-requirement RFP.	Reviewable draft produced in under 30 minutes.
TC-08	HITL	Reviewer clicks Approve.	Workflow advances to export; status changes to Approved.
TC-09	HITL	Reviewer edits a section inline, then approves.	Edit saved; final export reflects the human version; audit trail records the change.
TC-10	HITL	Reviewer sends back section 3 with a note.	Only section 3 regenerates using the note; rest of draft untouched.
TC-11	HITL	Draft has no human approval.	System blocks export. No response can be finalized without a human gate.
TC-12	Edge	Pricing API times out or is unavailable.	Workflow continues; price line flagged "missing/stale"; no fabricated number.
TC-13	Edge	A drafted claim has no supporting source.	Claim flagged as hallucination risk and surfaced to the reviewer.
TC-14	Edge	RAG finds no relevant content for a requirement.	Section marked "missing info" rather than filled with a guess.
TC-15	Edge	Corrupted or password-locked file uploaded.	Clear error message; user prompted to re-upload; no silent failure.

11 Tech stack & roadmap

Suggested stack

`LangGraph — orchestration` `LangChain — agent tooling`

`LLM (GPT / Claude) — drafting` `Vector DB (FAISS / Pinecone)` `Embeddings — retrieval` `FastAPI — backend` `React — frontend` `Pricing REST APIs` `PostgreSQL — audit log`

Future roadmap (post-capstone)

- Multi-language RFP support.
- Auto-suggested win themes based on past wins/losses.
- Team collaboration — multiple reviewers on one draft.
- Slack / email alerts when a flag needs an SME.
- Analytics: win rate vs. response quality over time.